

Renewable Energy Forecasting and ESG Implications

Analysis of Renewable Energy trends from 1960

Contents

Introduction	2
Data Analysis & Methodology.....	2
Data Sources & Cleaning.....	2
Modelling Approaches	3
ESG & DEIB Considerations	3
ESG Alignment	3
Fairness & Inclusivity Considerations	3
Key Findings	4
Country Wise Production	4
Incredible Rise in 70's	4
Analysis on few Select Countries	5
Hypothesis Tests around GDP, CO2 Emissions & Renewable Energy Production and Relationships between Continents	6
Modelling	7
Random Forest.....	7
Holt-Winters Model	8
Prophet Model.....	9
Linear Regression	10
Social & Environmental Impact	11
Conclusions	12
References	12

Introduction

One of the core moving parts of ESG (Environment, Social and Governance) is Renewable Energy. The transition from fossil fuels to renewable energy is something which began with the Oil Crisis of the 1970's where a special focus was placed onto renewable energy production. It is now a crucial agenda for policymakers, investors and businesses. This report and analysis aims to examine global trends from the 1960's using data from Kaggle (Renewable Energy (1960-2023), 2023), the World Bank (World Development Indicators | DataBank, n.d.) and Our World in Data (Ritchie & Roser, 2020).

The primary aim of this study is to analyze historical data & patterns and try to attempt predictive modelling to forecast future renewable energy trends. Leveraging statistical and machine learning techniques, this report focuses on these questions:

- 1) How has renewable energy production evolved over time? (Trends over time)
- 2) Can we predict future renewable energy production?
- 3) How do these trends impact global sustainability goals and ESG objectives?

Data Analysis & Methodology

Data Sources & Cleaning

The primary source of data is from Kaggle (Renewable Energy (1960-2023), 2023). It is a dataset which comprises data at a global scale with regards to renewable energy production (measured in kilotons of oil equivalent). To supplement/add on to this, additional data sources such as World Bank (World Development Indicators | DataBank, n.d.) [used for GDP purposes] and Our World in Data (Ritchie & Roser, 2020) [used for emission data]. These sources helped in integrating and exploring economic & environmental impact.

As is the case with every dataset available, cleaning of data is essential. Some key steps like renaming of columns, removing non-country aggregates (eg: In the Kaggle dataset, there were instances of aggregated data of regions available as rows which goes against

the motive of this analysis), filtering out data from certain countries (eg: Denmark and certain other countries had massive spikes in 1970 which couldn't be explained by any secondary level research, though, there was in fact the 1970's Oil Crisis (Oil crisis of the 1970s, n.d.)) and focusing our timeframe of modelling to be from 1990 to 2014 (the dataset has information from 1960 to 2015, however, 2015 data was misrepresented and didn't feel appropriate to be included). There was an additional identification of regional disparities which will be talked about in further detail later.

Modelling Approaches

During the process of selecting a model of forecasting, it felt appropriate to point out the suitability of each model under consideration:

MODEL	SUITABILITY	NOTES
Random Forest	Short-Term	Cannot extrapolate trends
XGBoost	Short-Term	Struggles with long-term
Linear Regression	Can't handle complex data	Non-linear trends
Holt-Winters	Suitable	Captures trends
Prophet	Suitable	Best for long-term

These approaches are talked about in detail in the upcoming sections.

ESG & DEIB Considerations

ESG Alignment

This study directly is supported by the Environmental pillar of ESG by:

- A) Tracking renewable energy production to assess global sustainability progress.
- B) Providing data-driven forecasts
- C) Identifying regional disparities

Fairness & Inclusivity Considerations

Certain low-income countries had missing data, which may introduce bias in forecasting. Couple of countries were excluded from the analysis due to misreporting/misrepresentation in the dataset obtained. Data from 2015 and before 1990 (only for forecasting) were ignored as they had either missing data or incredulous reporting (see notebook for the 1970 jump).

Key Findings

Country Wise Production

After combing through the data, the country with the most renewable energy produced was found to be China (which was largely expected). Countries like India and the U.S. closely followed them.

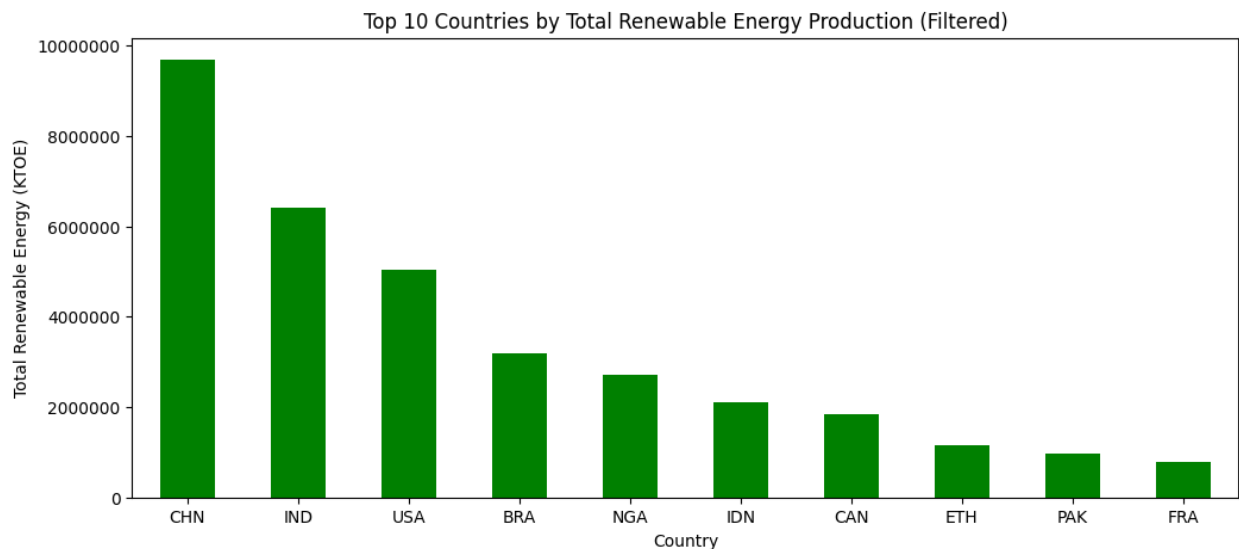


Figure 1: Top 10 Countries on Renewable Energy Production

Brazil (BRA) and Nigeria (NGA) are also in the top 5, which might be surprising given their developing economies. Brazil is known for its significant bioenergy production, particularly ethanol, which could explain its strong position (Brazil's biofuel production grows, reaching historical record, 2024).

Incredible Rise in 70's

"The 1973 oil embargo shook the global energy market. It also reset geopolitics, reordered the global economy, and introduced the modern energy era. The crisis and the iconic photographs of angry motorists fuming in gas lines are often evoked when oil and gasoline prices spike" (Lee, 2023). This was a convergence of three events: Geopolitical, Energy and a political crisis. To resolve the "energy crisis" and over reliance on oil from the Middle East which have been engulfed in war (to this day), countries began investing and ensuring that renewable energy is produced at a much higher rate than what was being done. The figure below clearly depicts the rise in production.

The onset of the 21st Century also came with it, technological prowess. Obviously, this is why the renewable energy production has seemed to have kept up with its exponential rise.

The reliance on renewable energy has also increased during this time and countries are very obviously spending vast amounts of money on research and development.

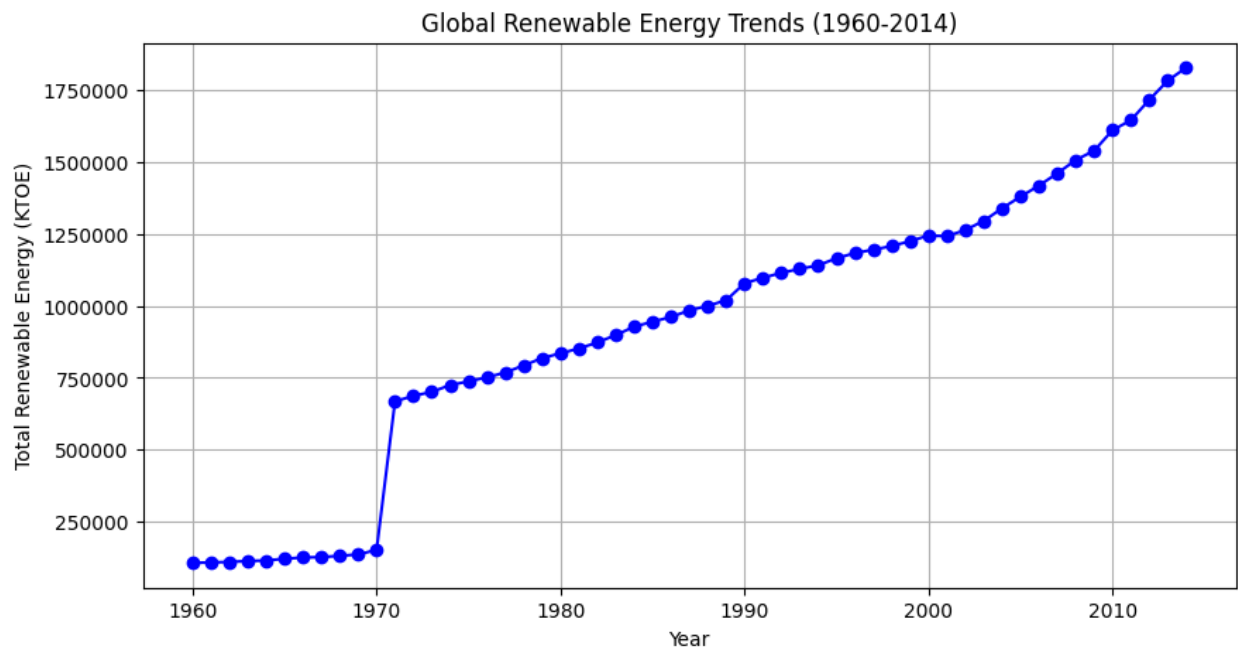


Figure 2: Rise in 1970s and Exponential Increase in the 21st Century

Analysis on few Select Countries

China has set ambitious goals for renewable energy expansion, which are reflected in its rapid growth in this sector (How China became the world's leader on renewable energy, n.d.). This can be seen in the graph below. Similar trends are observed with India and Brazil. India has overtaken the U.S. in production and Brazil has crept up to really close to the U.S. A strong national commitment to renewable energy, like China's, can result in substantial benefits for both energy security and environmental sustainability.

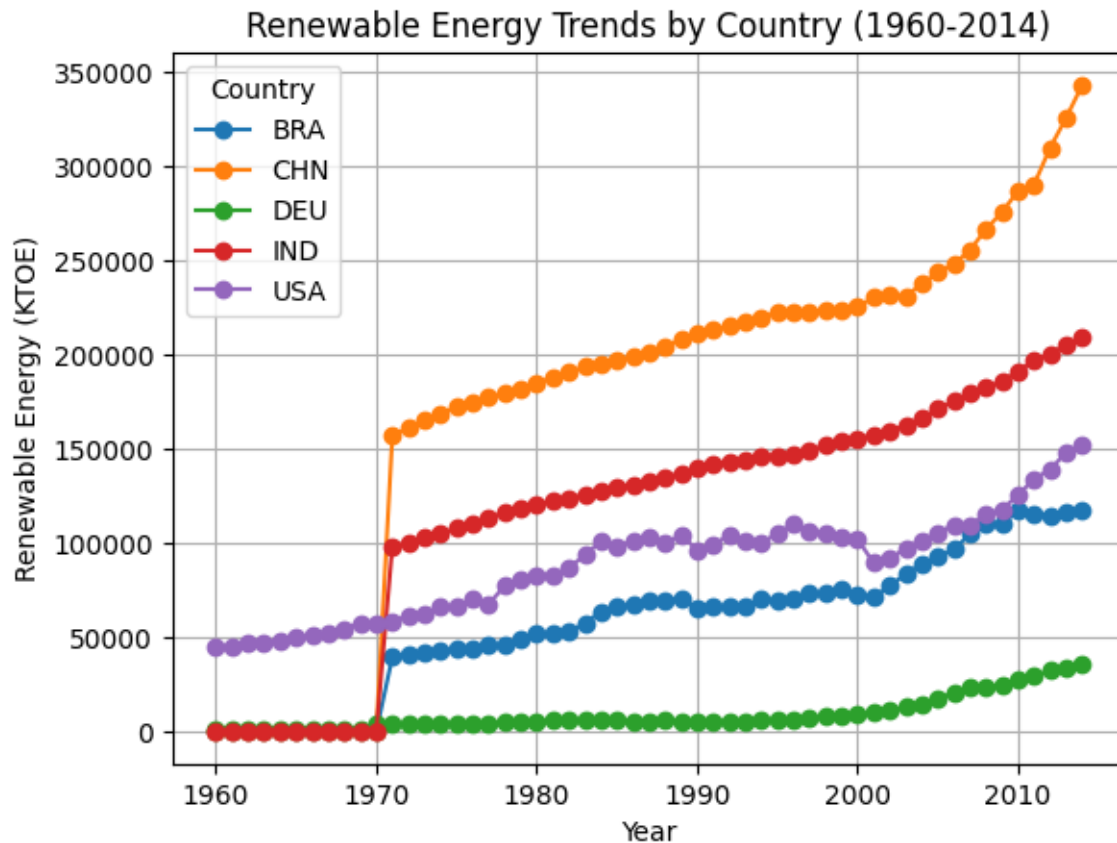


Figure 3: Rise of China's Renewable Energy Production

Hypothesis Tests around GDP, CO2 Emissions & Renewable Energy Production and Relationships between Continents

For the report, analysis was conducted on the relationship between GDP & Renewable Energy Production and CO2 emissions & Renewable Energy Production.

Firstly, the hypothesis was assumed to be that there is no relationship between GDP and Renewable Energy Production. However, with a strong correlation (score of 0.69) and very low p-Value (<1%), this hypothesis was rejected, and it can be concluded with strong confidence that there is a strong relationship between GDP & Renewable Energy Production.

Similarly, with an even higher correlation (score of 0.86) between CO2 emissions & Renewable Energy Production, it can be understood that CO2 emissions reduce with a higher production of renewable energy.

A continent-wise breakdown was also done, and it was found that wealthy continents like North America, Europe, and Asia use more renewable energy, while continents like Africa

and South America still have a lot of untapped potential. This points to the global disparity in renewable energy adoption and its intersection with economic development.

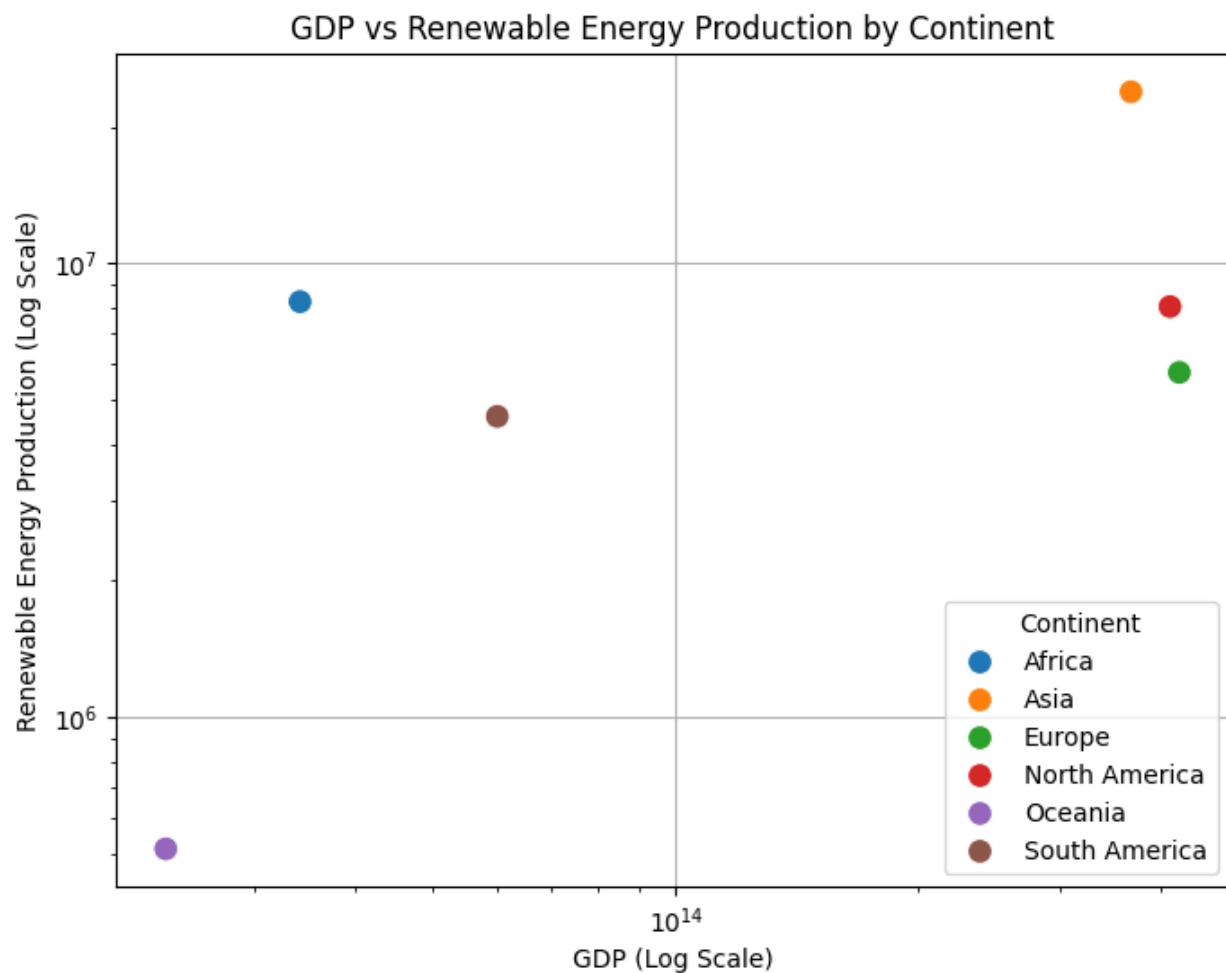


Figure 4: Richer the Continent, Higher the Renewable Energy Production

Modelling

Random Forest

Random Forest can be best explained with the example of consultation with multiple doctors for a lingering issue. Rather than just using one doctor's advice (a tree in this instance), it will be beneficial to consult multiple doctors to conclude (multiple trees → like combining wisdom of all the trees).

A Random Forest model was used to validate whether historical renewable energy production data exhibited a learnable and predictive pattern. The high correlation between actual and predicted values (from below) suggests that machine learning models can

effectively generalize past trends. As noted earlier, random forest is not an ideal way to predict long-term values, hence, this was used as a means of validating that similar models can learn and predict based on the data at hand.

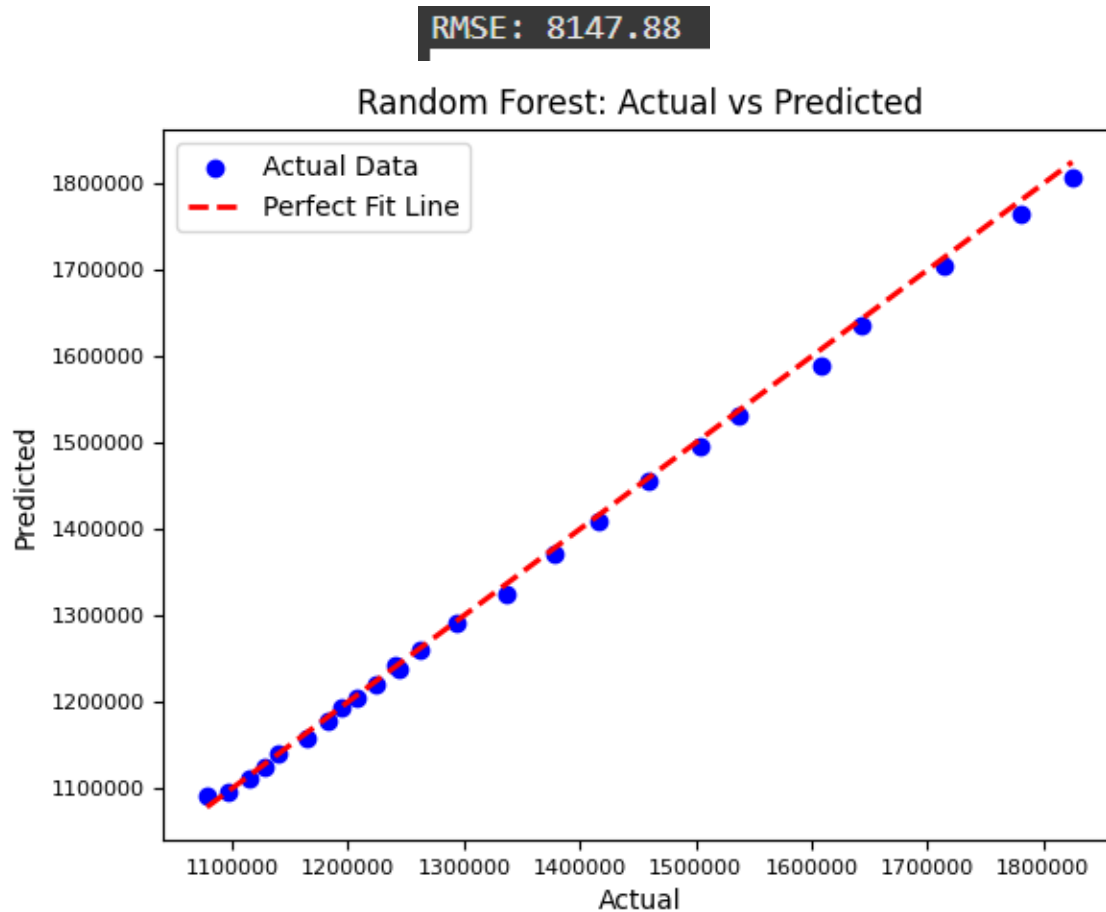


Figure 5: Random Forest can learn with our data

Holt-Winters Model

This model is generally known to look at general trends and predict outcomes in the future. Compared to other models like ARIMA (AutoRegressive Integrated Moving Average), Holt-Winters Model is more robust and best for data with trends such as the one with increasing renewable energy production.

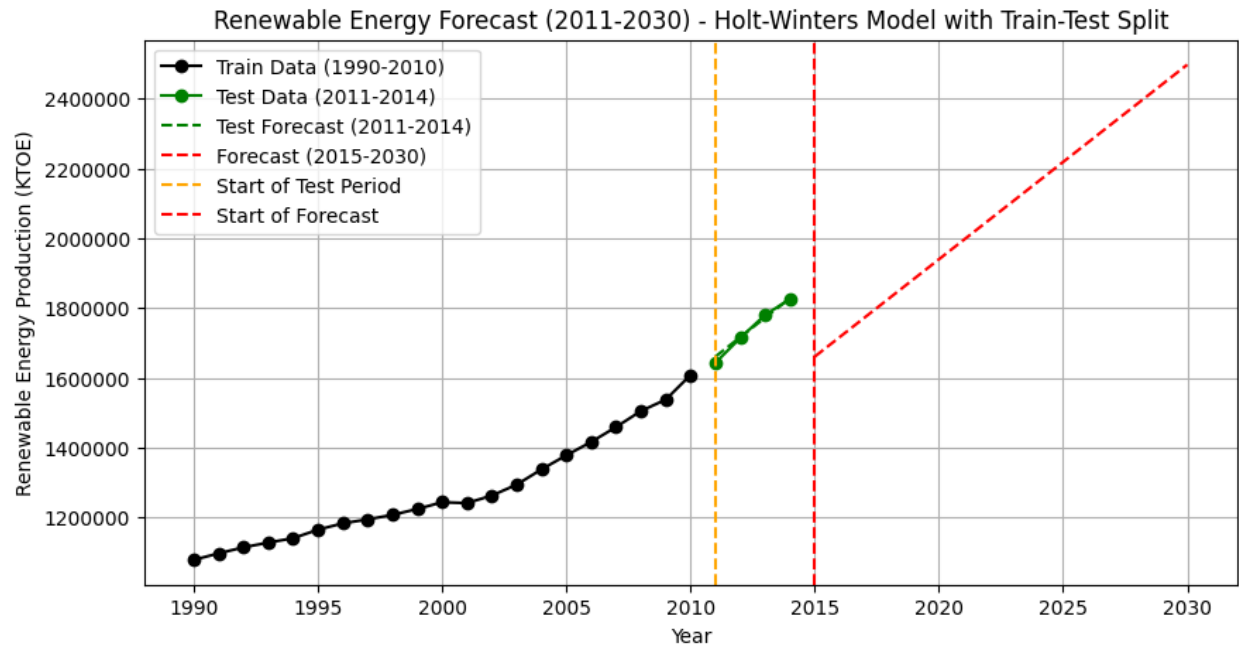


Figure 6: Holt-Winters model

By splitting the data into training and testing periods, the model ensures that predictions are reliable and not just based on historical trends. It is expected that the renewable energy production would reach a staggering 2,450,000 Kiloton Oil Equivalent. Let's now compare this with other modelling available.

Prophet Model

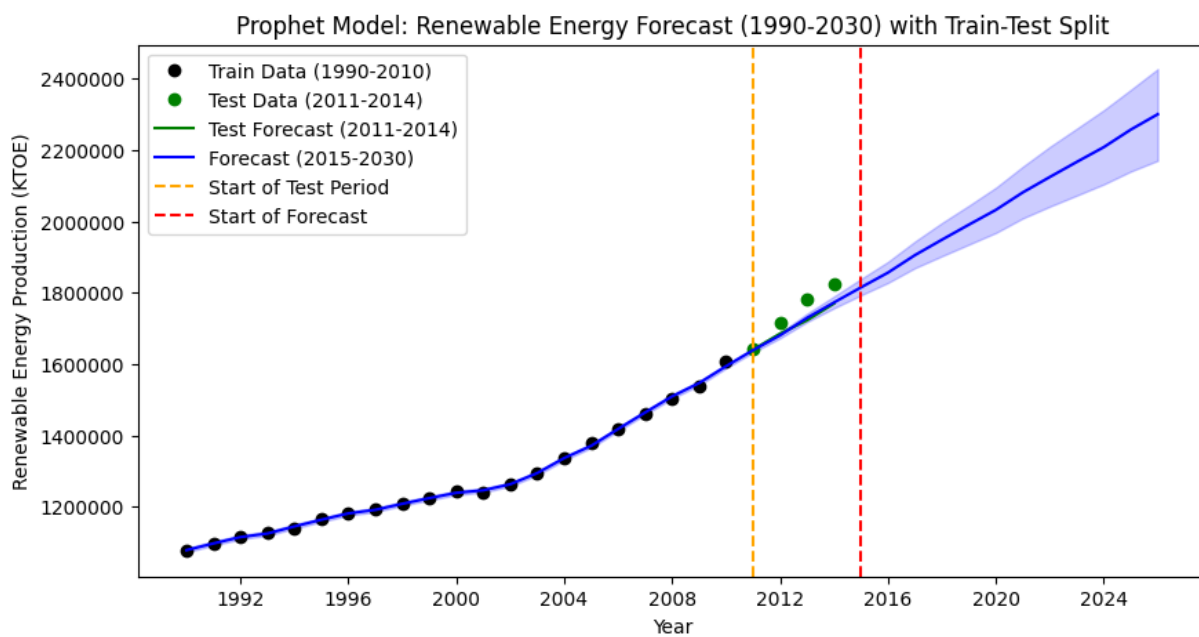


Figure 7: Predictions based on Prophet Modelling

Prophet Modelling can be referred to as a smart calendar model. It is very helpful to use when there is missing data and is very accurate in understanding how certain events affect future trends. The above graph displays/portrays an idea of the future whilst still displaying uncertainty (the shaded blue region). Hence, prophet modelling is a really good way to predict the future and by 2030 the renewable energy production can go up to 2,400,000 Kiloton Oil Equivalent.

Linear Regression

Linear regression is used to forecast future values based on relationship between two variables. Obviously, linear regression is subject to outliers the most and has to be taken into account when predictions are made.

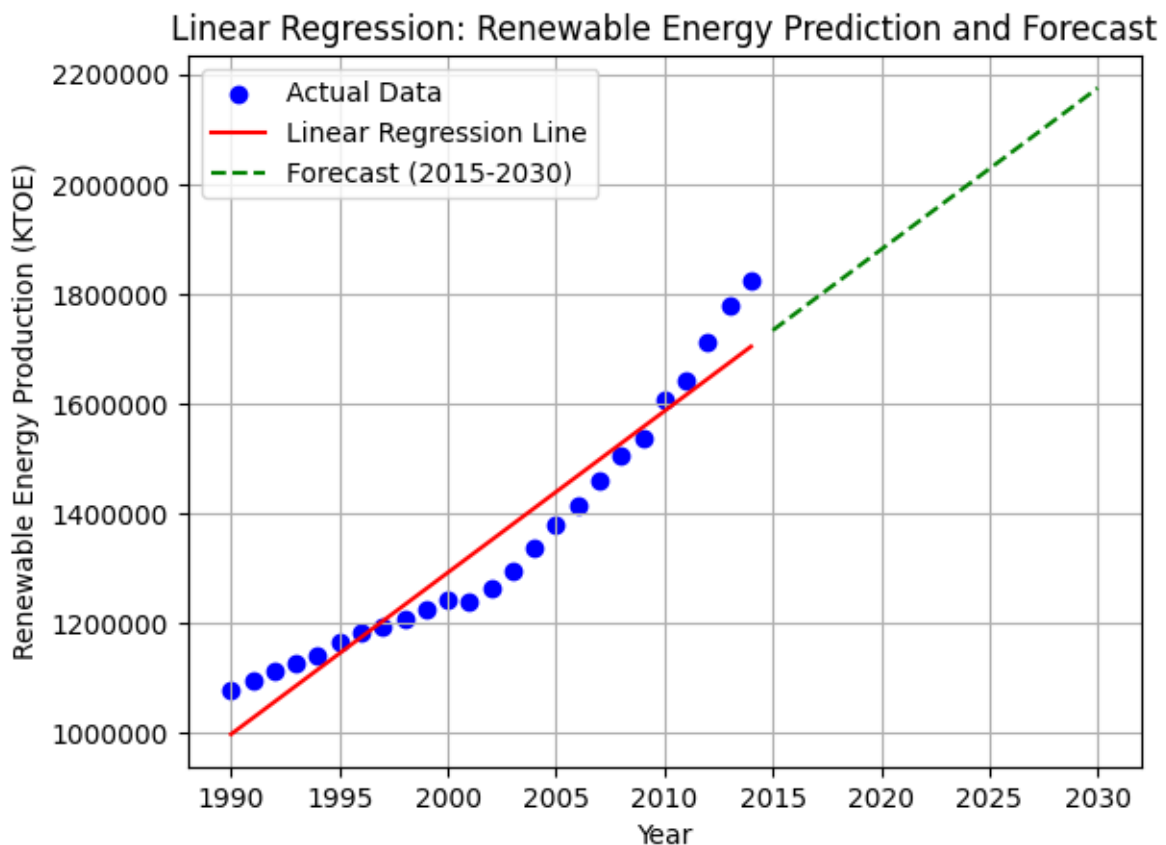


Figure 8: A Timid Prediction?

Linear regression seems to avoid the seemingly quadratic/exponential jump in the most recent years and looks to indicate that by 2030, the production would reach only 2,180,000 Kiloton Oil Equivalent. This seems to be an obvious under-prediction. Maybe another method like Quadratic regression would be a better fit here (Fig below).

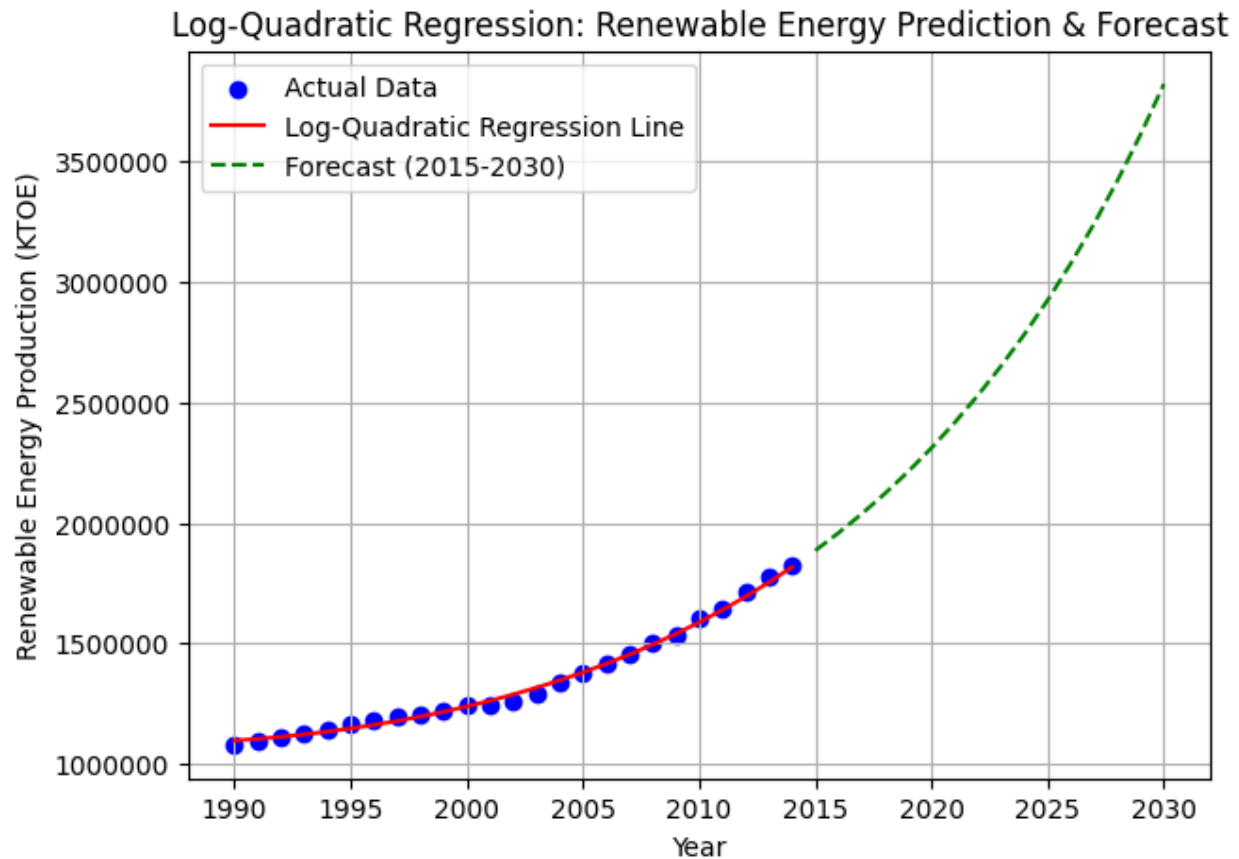


Figure 9: Over-Estimation??

Despite fitting really well with the current data, this seems to be an overly optimistic forecast where it is predicted that by 2030, a mind boggling 3,650,000 Kiloton Oil Equivalent of renewable energy would be produced by the world (like 1 million greater than other predictors).

Social & Environmental Impact

Our findings indicate significant growth in renewable energy production over the past three decades. This trend is expected to continue, but challenges remain. Countries with resource limitations will struggle to expand into this sector as this is still a very expensive industry to expand into.

Government policies, subsidies and technological advancements would be key in ensuring growth in this sector as seen in the 21st century boom (technology) and 1970s explosion (policy).

The forecasts here enable actionable insights for policymakers, investors and energy companies to speed up their transition to clean, green and sustainable energy.

Conclusions

The report shows that whilst renewable energy is a growing industry, just like all industries, there will be growing pains. Regional disparities and economic barriers being the biggest hurdle towards “green” energy. The predictive models performed here offer a long-term vision for growth which could help guide policy and investment. By leveraging data analytics for renewable energy forecasting, we can contribute to a more sustainable and equitable energy future.

References

- Brazil's biofuel production grows, reaching historical record.* (2024, July 23). Retrieved March 1, 2025, from Secretaria De Comunicação Social: <https://www.gov.br/secom/en/latest-news/2024/07/brazil2019s-biofuel-production-grows-reaching-historical-record>
- How China became the world's leader on renewable energy.* (n.d.). Retrieved March 1, 2025, from Yale E360: <https://e360.yale.edu/features/china-renewable-energy>
- Lee, K. (2023, October 16). *The 1973 oil Crisis: Three crises in one—and the lessons for today.* Retrieved March 1, 2025, from Center on Global Energy Policy at Columbia University SIPA | CGEP: <https://www.energypolicy.columbia.edu/publications/the-1973-oil-crisis-three-crisis-in-one-and-the-lessons-for-today/>
- Oil crisis of the 1970s.* (n.d.). Retrieved March 1, 2025, from Energy Education: https://energyeducation.ca/encyclopedia/Oil_crisis_of_the_1970s
- Renewable Energy (1960-2023).* (2023, September 13). Retrieved February 27, 2025, from Kaggle: <https://www.kaggle.com/datasets/imtkaggleteam/renewable-energy-1960-2023/data>
- Ritchie, H., & Roser, M. (2020, June 10). *CO₂ emissions.* Retrieved February 28, 2025, from Our World in Data: <https://ourworldindata.org/co2-emissions>
- World Development Indicators | DataBank.* (n.d.). Retrieved February 28, 2025, from World Bank: <https://databank.worldbank.org/source/world-development-indicators#>